MHNet: A Masked Hybrid Network for Robust Water Body Segmentation From Aerial Images

Shuo Wang, Bin Wei, Boneng Shi, Ninglian Wang, Yuzhu Zhang, and Yan Zhu

***Abstract*— Accurate segmentation of water bodies from aerial images is critical for advancing our understanding of climate change, improving flood prevention and mitigation efforts, and supporting ecological monitoring. Recently, deep learning-based methods have made remarkable progress in water body segmentation. However, there still exist a series of challenges in practical applications, including mis-segmentation of low-contrast regions, difficult delineation of complex terrain boundaries, and loss of small water features. While most of the existing methods are designed for supervised learning with paired samples, they may also benefit from self-supervised learning techniques such as masked autoencoders (MAE). In this work, we focus on the water segmentation problem and propose a new water segmentation framework, named MHNet. MHNet integrates a hybrid multiscale encoder–decoder network, combining convolutional and transformer-based components to effectively capture global context while minimizing computational costs. A key innovation is the adaptation of the MAE mechanism, where masks are applied to multiscale Restormer block outputs during the training phase, enabling the model to better integrate local and global information, thereby enhancing boundary segmentation accuracy. Additionally, we propose a multichannel feature fusion (MCFF) module that synthesizes masked feature maps across scales, reducing redundancy and improving generalization by capturing both fine details and contextual information. Extensive experiments on multiple public datasets demonstrate that MHNet outperforms state-of-the-art methods, highlighting its effectiveness and robustness in water body extraction tasks. MHNet is deployed and performs online predictions via Google's Vertex AI platform, thereby integrating it into a Geographic Information System (GIS) using Google Earth Engine (GEE) for accurate and efficient extraction of lakes on the Tibetan Plateau.**

***Index Terms*— Deep learning, lake water body extraction, mask autoencoder, semantic segmentation.**

I. Introduction

LAKE water body segmentation is a key focus in remote sensing and image analysis, aiming to segment water areas from surrounding landscapes in satellite and aerial imagery. This technology is essential for understanding aquatic ecosystems, hydrological processes, and environmental dynamics. Changes in water bodies, such as formation, expansion, contraction, or disappearance, have profound implications for ecological balance, regional and global climate systems, and the carbon cycle. Therefore, rapid and accurate extraction of water body information is essential for environmental conservation, disaster mitigation, water resource management, and agricultural planning, enabling timely and informed decision-making in response to dynamic environmental conditions.

In recent years, deep learning has significantly advanced the segmentation of lake water bodies from aerial images, particularly through applying convolutional neural networks (CNNs) and fully convolutional networks (FCNs). CNNs [1], [2], [3], renowned for their robust feature extraction capabilities, excel in handling complex patterns and structures in remote sensing images. Meanwhile, state-of-the-art semantic segmentation frameworks are predominantly based on FCNs [4], [5], [6], leveraging end-to-end pixel-level classification to effectively identify water bodies in aerial imagery. Substantial research efforts have been devoted to applying these deep learning methods to remote sensing image segmentation. However, CNNs are inherently limited by their receptive field, which restricts their ability to capture long-range pixel dependencies. A widely adopted solution involves multiscale feature fusion strategies. To address the need for multiscale feature extraction, researchers have developed advanced CNN-based methods, such as PSPNet [7], and DeepLab [8]. These approaches further advance the development of semantic segmentation networks by benefiting from the global modeling capabilities of pyramid pooling module.

More recently, Dosovitskiy et al. [9] proposed vision transformer (ViT), which successfully introduced the self-attention mechanism in Transformers into the visual field. This mechanism calculates pixel-level responses by weighting all other positions in an image, enabling a more global and flexible representation. On the one hand, the single-input–single-output architecture of ViT ignores the potential explicit information from multiple image scales and lacks the ability to explore the complementary implicit information between different scales. On the other hand, the training of the ViT model requires a large amount of data to compensate for its little inductive bias to achieve best performance and generalization ability. Additionally, the computational complexity of the multihead self-attention (MSA) mechanism increases quadratically with image resolution, requiring researchers to strike a balance between model expressiveness and computational efficiency. To mitigate these challenges, hybrid architectures combining

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convolutional and transformer components [10], [11], [12], [13] have been developed. These models integrate the local feature extraction strength of CNNs with the global modeling capabilities of ViTs, producing hierarchical feature representations. This approach reduces computational overhead while preserving accuracy, achieving state-of-the-art performance in image classification, object detection, and semantic segmentation.

Although existing methods have indeed achieved impressive performance, they still face several challenges in their application to the water body segmentation task. First, the scarcity of large-scale pixel-level annotated datasets tailored for water segmentation tasks directly limits the generalization ability of these methods in different scenarios. Furthermore, existing methods often experience issues with incorrect segmentation when dealing with areas where water bodies and land have low contrast, such as the surfaces of lakes covered with ice and abnormal water surfaces, for example, water surfaces with bubbles. Therefore, there is an urgent need to prioritize the development of methods that reduce dependence on paired data while establishing robust representations with good generalization capabilities to adapt to the segmentation requirements of abnormal water bodies. What is nice is that, the focus of visual representation learning has increasingly shifted from supervised learning with labeled datasets [14], [15], [16], [17] to self-supervised pretraining using pretext tasks [18], [19], [20], [21]. Among self-supervised approaches, masked autoencoders (MAEs) [22] have emerged as a prominent method by extending masked language modeling techniques to the visual domain. The MAEs reconstruction capability helps mitigate the data scarcity challenges often faced by ViTs, while also enhancing training efficiency and reducing the risk of overfitting.

Combining the advantages and challenges of the aforementioned deep learning methods, this article proposes a novel multiscale encoder–decoder network, named MHNet, for segmenting water bodies from aerial images. MHNet integrates a hybrid convolution-transformer architecture to overcome the limitations of traditional CNNs and improve the accuracy of water body feature extraction and boundary segmentation. First, MHNet employs a multiscale progressive encoder–decoder architecture based on the ViT. This design addresses the restricted receptive field of CNNs, enabling the model to capture features across various scales and enhancing its ability to identify water body boundaries. Second, given the high computational cost of the MSA mechanism in ViTs, MHNet incorporates multi-Dconv head transposed attention (MDTA) and gated-Dconv feed-forward network (GDFN) modules inspired by Restormer [23]. These modules reduce computational overhead while maintaining the model's ability to extract meaningful features. Additionally, MHNet introduces a self-supervised learning mechanism inspired by MAEs to address the challenges of limited labeled datasets. By applying masks to the output feature maps of the multiscale Restormer blocks, the model improves its learning efficiency and generalization ability for water body feature representation. To further address the issue of information loss caused by directly masking input images, a multichannel feature fusion (MCFF) module is introduced. The MCFF module integrates masked token information with local and global features at different scales, achieving effective multiscale feature fusion. This approach enhances the model's capability to delineate water body boundaries accurately and capture fine-grained details.

In summary, this article makes several key contributions.

1) *Hybrid multiscale encoder–decoder network:* A novel multiscale encoder–decoder architecture, MHNet, is introduced. By hybridizing convolutional and transformer-based components, the model incorporates global context into feature extraction and reduces the computational overhead of the transformer's attention mechanism.
2) *Integration of MAE mechanism:* MHNet adapts the concept of MAEs by applying masks to the output feature maps of multiscale Restormer blocks. This design allows the model to effectively integrate local and global information, enabling multiscale feature fusion and improving the segmentation accuracy of water body boundaries.
3) *MCFF module:* A novel MCFF module is proposed to synthesize masked feature maps across different scales, reducing feature redundancy and enhancing the generalization ability of MHNet. This module enables the model to better capture both fine details and contextual information, significantly improving segmentation performance.
4) *Deployment of MHNet and integration with Geographic Information System (GIS) systems:* Deploying MHNet on Google Cloud with the Vertex AI platform for online inference and integration with Google Earth Engine (GEE) for accurate and efficient water body segmentation demonstrates the scalability of MHNet for large-scale satellite image segmentation tasks.

II. Related Work

A. Lake Water Body Segmentation

With the development of computer vision technology, a large number of excellent works have emerged in the field of lake water segmentation. These methods can be divided broadly into two categories: methods based on traditional computer vision and methods based on deep learning.

1) Methods Based on Traditional Computer Vision: Shen and Li [24] utilized the Adaboost algorithm to combine multiple weak classifiers based on spectral information, water indices, and spectral relationships to form a strong classifier for water body extraction. Huang et al. [25] proposed a comprehensive threshold method for water body extraction based on MODIS data, which integrates several factors such as normalized difference vegetation index (NDVI), spectral characteristics of water bodies, cloud shadows, and SRTM digital elevation information. McFeeters [26] introduced a new method named normalized difference water index (NDWI), which makes use of reflected near-infrared radiation and visible green light to better distinguish between water bodies and other features. Mishra and Prasad [27] developed an

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automatic perceptron model using feature vectors that include low short-wave infrared reflectance (SWIR), band reflectance comparisons, and modified NDWI to extract water bodies from Landsat images.

2) Methods Based on Deep Learning: Feng et al. [28] proposed an enhanced deep convolutional encoder–decoder (DCED) network tailored to water body extraction, incorporating superpixel segmentation and conditional random fields (CRFs) to improve segmentation precision. Fu et al. [29] introduced a weakly supervised feature-fusion network to adapt to the unique characteristics of objects in remote sensing images. Wang et al. [30] introduced MSLWENet, which combines multiscale feature extraction and dense connection modules to enhance the accuracy of small water body detection and suppress noise interference in Google Remote Sensing Images. Liu et al. [31] proposed a Siamese network, which enhances the reliability of water body extraction from C-band SAR images by integrating polarized VV/VH data with digital terrain model (DTM), digital surface model (DSM), and land use information, thereby mitigating the impact of backscatter sensitivity. Zhong et al. [32] modeled the key features that are distinguishable and suitable for segmenting lake water by analyzing the difference in feature representation between lakes and other ground objects. Addressing the sparsity of contextual priors, Xiang et al. [33] developed DensePPMUNet-a, which integrates dense global contextual priors with attention mechanisms and Res Path to improve segmentation performance. Chen et al. [34] utilized an adaptive clustering algorithm for superpixel segmentation and a CNN architecture for high-level feature extraction for the extraction of urban water bodies from high-resolution remote sensing (HRRS) imagery. Sun et al. [35] proposed WaterDeep, which integrates Xception baseline network for low-level water feature extraction, and harnesses densely connected atrous spatial pyramid pooling (ASPP) modules to assimilate multiscale data into sophisticated high-level features. Zhong et al. [36] introduced a multiscale encoder–decoder structure information enhancement network with two-way channel attention mechanism, and a deep dilation residual structure is used in the encoder module. Wang et al. [37] presented HA-Net, which is a multiscale structure that combines the features of ResNet-34 with a feature pyramid network and adaptive weights are distributed to global information using the hybrid-scale attention block. Liu et al. [38] proposed a double multiplication attention module UNet (DMAM-UNet) modified from the UNet network. Wang et al. [39] proposed a new method by combining GEE with a multiscale CNN (MSCNN) to extract urban water from Landsat images. Xu et al. [40] proposed an information-expanding network IE-Unet based on the traditional network U-net, where the rotation equivariant convolution, rotation-based channel attention mechanism, and the optimized batchnorm are adopted jointly. Wang et al. [41] proposed SADA-Net for HRRS images water bodies extraction, which considers multiscale information, context dependence, and shape features and integrates shape feature optimization (SFO), ASPP, and dual attention modules. Yuan et al. [42] introduced MC-WBDN, a deep CNN (DCNN)-based methods, which incorporates a multichannel fusion module, an enhanced ASPP module, and space-to-depth/depth-to-space operations to outperform water body detection. Researchers have also attempted to apply ViTs to water body segmentation tasks. For example, Qi et al. [43] combined an improved geodesic active contour (GAC) model with ViTs, introducing a differentiable spectral clustering (DSC) module to enable automatic contour initialization and improve fine-grained water body extraction. These works collectively demonstrate the evolution of deep learning methods toward more accurate and context-aware approaches for lake water segmentation in remote sensing.

B. Masked Autoencoders

As one of the popular masked image modeling (MIM) methods, MAEs [22] stands out for its computational efficiency since it only feeds visible patches into encoders. In the MAE architecture, images are divided into nonoverlapping patches, and a high proportion of these patches (e.g., 75%) are masked. The remaining visible patches are processed by an asymmetric encoder, which learns contextual relationships, while a lightweight decoder reconstructs the original image. The masking strategy significantly reduces computational overhead during pretraining by requiring the encoder to process only a small fraction of the input.

MAEs have been shown to effectively adapt ViTs to various tasks by enabling the model to capture semantic information. This simple yet efficient framework provides a scalable solution for visual representation learning, particularly in scenarios with limited labeled data. Zhou et al. [44] combined the Unet decoder with the MAE pretrained ViT encoder to achieve excellent medical image segmentation performance. Reed et al. [45] proposed scale-MAE, which introduces scale invariance into MAE encoders. Scale-MAE exhibits excellent building segmentation performance on remote sensing datasets with multiple scale and resolution characteristics. The ability of MAEs to enhance feature representation offers promising potential for tasks such as water body segmentation.

Based on the development of the above state-of-the-art techniques, this work proposes a new hybrid framework to address the challenges of fine-grained, multiscale lake water body segmentation in remote sensing applications.

III. Proposed Method

The proposed MHNet is a multiscale hybrid convolution-transformer model designed for water body segmentation tasks. It incorporates a multiscale encoder–decoder architecture, a masking strategy, and a multiscale fusion mechanism. The overall architecture is illustrated in Fig. 1. This section comprehensively describes the network design and key components, highlighting their functionality and integration.

A. Network Architecture Overview

The model begins by embedding the input image into patches using a convolutional embedding operation. Unlike the ViT, which relies on a fully connected (FC) layer for embedding, a convolutional layer is adopted here to improve computational efficiency and better exploit local spatial information. By adjusting the convolution kernel size and stride,

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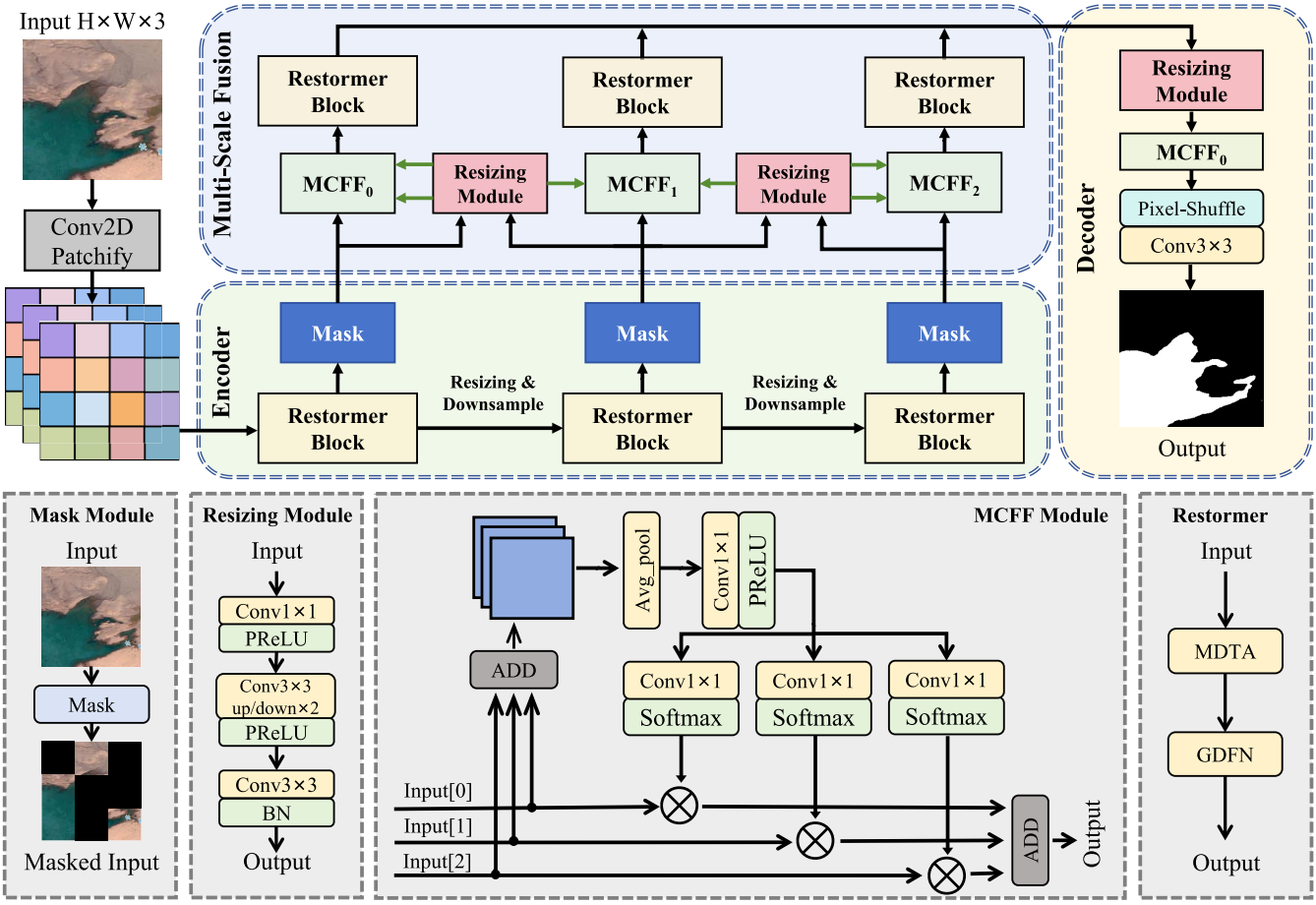


Fig. 1. Overview of the proposed MHNet. The input of the network is water body image and the output is segmented water body image. It should be noticed that the mask is only used at the training stage. MDTA and GDFN stand for MDTA and gated-Dconv feedforward network, respectively.

the embedding operation allows control over patch size and overlap, achieving both efficient computation and effective feature extraction [46]. The embedding process is expressed as

$$F_{\text{embed}} = \text{BN}\big(\text{LeakyReLU}\big(\text{Conv}_{s,p}(I, K_h)\big)\big) \tag{1}$$

where Conv denotes for convolution operation, s the stride, p the padding, and K_h the kernel size, and I the input image tensor. Batch normalization (BN) and LeakyReLU activation are applied to enhance stability and representation quality.

The embedded patches (F_{embed}) are then processed by the multiscale encoder, where a masking strategy is applied to feature maps at different scales. These processed features are fed into the MCFF module for effective multiscale integration. The decoder subsequently reconstructs the segmented water body image by fusing features across scales, restoring spatial resolution, and learning residual information. Sections III-B–III-D provide detailed descriptions of the key components of the proposed method.

B. Multiscale Encoder–Decoder

The multiscale encoder–decoder structure underpins the core functionality of MHNet. The encoder begins with the embedded patches (F_{embed}) and progressively extracts hierarchical features using Restormer blocks. The outputs of these blocks (F_r^n) are further processed through spatial-to-depth transformations and downsampling operations to produce multiscale features F_{rsd}^n. These downsampled features serve as inputs for subsequent Restormer blocks at deeper scales. The encoder operations are summarized as

$$\begin{aligned} F_{rs}^n &= \text{space_to_depth}\big(F_r^n\big) \\ F_{rsd}^n &= \text{resize_down}\big(F_{rs}^n\big) \end{aligned} \tag{2}$$

where resize_down represents the downsampling operation. Masking is applied at each scale to the feature maps F_{rsd}^n, further refining the extracted features.

The decoder reconstructs the segmented water body image by aggregating the multiscale feature maps. First, the fused feature maps are resized back to the original input resolution using a resizing module. To recover spatial details, a pixel-shuffle operation followed by a 3×3 convolutional layer transforms the fused features into the reconstructed image. Residual learning is employed by adding the upsampled input image to the final output. The decoding process is formulated as

$$\begin{aligned} \mathbf{F}_{de} &= f_{\text{MCFF0}}\big(f_{Rz}\big(\mathbf{F}_{f1}, \mathbf{F}_{f2}, \ldots, \mathbf{F}_{f\text{depth}-1}\big); \mathbf{W}_0\big) \\ \mathbf{Z} &= \text{Conv}(P_s(\mathbf{F}_{de})) \end{aligned} \tag{3}$$

where Z is the water body segmentation result, F_{fn} is the output from multiscale fusion, $P_s()$ represents the pixel-shuffle

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operation, and W_0 denotes the learnable weights for feature fusion, at the same time f_{MCFF_0} and f_{Rz} represents MCFF_0 module and resizing module.

C. Masking Strategy

MHNet introduces a novel masking strategy applied to the output feature maps of the multiscale encoder, rather than directly applies the mask strategy to the input image as in traditional MAE. In this approach, a certain proportion of patches along the feature-wise dimension are randomly replaced with zeros or random noise instead of being discarded. This design achieves two objectives: 1) it reduces interchannel redundancy by creating masked tokens in the channel dimension and 2) it facilitates the exploration of rich feature information through the subsequent self-attention module. By masking input feature maps, the model learns to infer spatial–spectral relationships from context, enhancing generalization across complex scenarios while reducing reliance on labeled data. This strategy enables robust water body segmentation under challenging conditions, with empirical results confirming its effectiveness despite perceived noise introduction.

Here, the masking rate is kept low to balance the increased difficulty of reconstruction introduced by masking, and a multiscale encoder–decoder network is employed to enhance token utilization. Additionally, dropout within the Restormer blocks complements the masking strategy, further improving generalization. The masking process is defined as

$$F^n_{\mathrm{mask}} = \mathrm{Mask_to_token}\left(\mathrm{Random_Mask}\left(F^n_{\mathrm{in}}\right), T\right) \quad (4)$$

where F_{mask} and F_{in} represent the masked and input feature maps at scale n, respectively, and T denotes a learnable patch initialized with random noise.

D. Modules in Multiscale Fusion

The multiscale fusion process integrates three key modules: resizing modules, the MCFF module, and Restormer blocks.

1) Resizing Module: The resizing module is designed to adjust the resolution of feature maps at different scales. Given that the feature maps are masked, suitable sampling functions are required to process the complex information effectively. After evaluating various interpolation methods and their associated computational overhead, the single convolution/transposed convolution sampling technique was selected to balance performance and efficiency.

It should be noted that a single resizing module can only perform 2x upsampling or downsampling. Therefore, for larger scaling factors (e.g., 8× upsampling), multiple resizing modules are stacked. The detailed structure of the resizing module is shown in Fig. 1.

2) MCFF Module: The MCFF module is designed to fuse multichannel feature maps generated by the encoder. Specifically, the module receives three input branches: one containing the masked feature map output by the Restormer block at the corresponding scale, and the other two containing feature maps resized by the resizing module. The MCFF module performs MCFF, where the fusion strategy is controlled by a neural network that learns the optimal way to combine these feature maps, effectively reducing redundancy between channels.

In MHNet, the MCFF_0 module of the encoder shares parameters with the MCFF_0 module in the decoder. This ensures a degree of consistency between input and output feature processing, allowing for better feature reuse and enhancing the model's overall performance.

a) Restormer block: The Restormer block [23] is used for feature extraction and consists of two key components: MDTA and GDFN. The feature extraction process is defined as

$$F_{\mathrm{out}} = f_{\mathrm{GDFN}}(\mathrm{LN}(f_{\mathrm{MDTA}}(\mathrm{LN}(F_{\mathrm{in}})))) \quad (5)$$

where LN represents layer normalization, and f_{MDTA} and f_{GDFN} refer to the respective MDTA and GDFN modules. This structure facilitates efficient feature extraction by leveraging attention mechanisms and feedforward networks, leading to enhanced model performance.

In general, the resizing module and MCFF module are used for multiscale feature fusion, and the Restormer block is used to maintain the excellent representation learning ability of the transformer while reducing the computational complexity of the model.

E. Loss Function

The training process is guided by a hybrid loss function that combines binary cross-entropy loss (BCELoss) and dice loss, enabling the model to balance pixel-level classification accuracy and spatial coherence in segmentation tasks.

1) Binary Cross-Entropy Loss: BCELoss is particularly effective for pixel-level classification tasks, addressing class imbalance by measuring the difference between predicted and actual probability distributions at the pixel level. The BCELoss function is defined as

$$L_{\mathrm{BCE}}(x, y) = \frac{1}{N}\sum_{i=1}^{N}\left(\log\left(1 + e^{-x_i}\right) + (1 - y_i)x_i\right) \quad (6)$$

where N is the batch size, x_i the predicted probability, and y_i the ground truth label.

This is particularly important given the class imbalance in remote sensing images, where water bodies often occupy a small fraction of the total pixels. However, BCE Loss alone may struggle to capture the overall structure and shape of water bodies, especially in cases of irregular or fragmented water regions. To address this, we incorporated dice loss.

2) Dice Loss: The dice loss is based on the dice coefficient, which measures the overlap between predicted and true regions. It accounts for regional dependencies by calculating the loss for each pixel in the context of its surrounding pixels, making it particularly useful for accurately extracting small objects and delineating boundaries. The dice loss is defined as

$$L_{\mathrm{Dice}} = 1 - \frac{1}{N}\sum_{n=1}^{N}\frac{2\sum_{i=1}^{M} p_{ni}\, y_{ni} + \varepsilon}{\sum_{i=1}^{M} p_{ni} + \sum_{i=1}^{M} y_{ni} + \varepsilon} \quad (7)$$

where p_ni and y_ni are the predicted and true values for pixel i in sample n. The term ε is a smoothing factor added to avoid division by zero.

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3) Combined Loss Function: To achieve a balance between classification accuracy and spatial coherence, the total loss function is defined as

$$L_{\text{total}} = \alpha \times L_{\text{BCE}} + \beta \times L_{\text{Dice}} \tag{8}$$

where α and β represent the combination coefficients of L_{BCE} and L_{Dice}, respectively. Through extensive experimentation, we found that setting $\alpha = 0.5$ and $\beta = 1.0$ provides the best balance between the two loss components, leading to improved model performance. More experimental analyses are shown in Section IV. This hybrid loss function guides the model to learn both pixel-wise precision and the broader spatial structure of the segmented regions, leading to improved performance in water body segmentation tasks.

IV. Experimental Results and Analysis

In this section, we first describe the experimental datasets. Next, we outline the experimental settings, including implementation details, optimization configurations, and evaluation metrics. We then present the experimental results of the proposed MHNet, comparing its performance with state-of-the-art water body extraction methods. Furthermore, we conduct a series of ablation experiments to validate the contributions of the individual components of our method and discuss its limitations. Finally, we provide an example of MHNet's cloud deployment and integration with GIS systems.

A. Datasets

In this study, we utilize five distinct datasets to evaluate the performance of the proposed model. In order to achieve the best performance of the model, 70% of the data was selected for training after comprehensively considering the size of the dataset, and the remaining 30% was used for testing.

1) Tibetan Plateau Dataset: The Tibetan Plateau dataset (TPD), created by Wang et al. [30] at Lanzhou University, comprises 6,774 RGB images, each with dimensions of 256 × 256 pixels, a depth of 24 bits, and a resolution of 96 DPI. This dataset focuses on lakes, excluding other water bodies such as rivers, reservoirs, and ponds. Most lakes have clear boundaries and are often surrounded by desert or Gobi landscapes. The images come from Google Earth and cover the entire Tibetan Plateau. We used 4,742 images for training and 2,032 for testing.

2) Gaofen Image Dataset: The Gaofen Image dataset (GID) [47] is a large-scale land cover dataset collected using the Gaofen-2 satellite, offering extensive coverage, diverse distribution, and high spatial resolution. The GID dataset contains elements such as urban buildings, forests, farmlands, and reservoirs. It is also the largest dataset in this study. GID includes 150 pixel-level annotated images, each sized 6800 × 7200 pixels. These high-resolution images were split into 256 × 256 blocks, generating 31 500 images. Among these, 22,050 images were used for training, and 9,450 images were allocated for testing.

3) WHDLD: The WHDLD dataset [48] is a dense labeling dataset that includes 4,940 RGB images with a resolution of 256 × 256 pixels. The WHDLD dataset was chosen to complement the TPD dataset due to its rich inclusion of various water features, notably ponds and reservoirs. It is particularly suitable for multilabeling tasks such as remote sensing image retrieval (RSIR) and semantic segmentation. In this study, we used 3,458 images from the WHDLD dataset for training and 1,482 images for testing.

4) DLRSD: The DLRSD dataset [49] is based on the UC Merced LandUse dataset, with images from the USGS National Map. It supports deep learning research in remote sensing, especially for pixel-level tasks like semantic segmentation. The dataset has 21 land cover classes, each with 100 RGB images sized 256 × 256 pixels. To test MHNet's generalization, we used all 100 images from the "river" category solely as the test set.

5) ISIC2018 Dataset: To evaluate the generalizability of MHNet, the International Skin Imaging Collaboration (ISIC) 2018 dermatoscopic image dataset [50] for skin lesion analysis toward melanoma detection which is published by the ISIC, whose task 1 focuses on the lesion segmentation task and contains 2,594 dermatoscopic images and their corresponding segmentation masks, was selected. The morphology and texture of skin lesions are complicated which may include a variety of features such as pigment networks, blobs, and streaks. These features are represented in various forms in the image, which increases the difficulty of segmentation. For example, some lesions may contain multiple subregions, or the lesion area may have low contrast with the surrounding skin.

This study focuses on the analysis of RGB imagery synthesized from multisource remote sensing data. The experimental datasets integrate imagery from various platforms, including the Gaofen-2 satellite, UAV-based aerial photography, and Google Earth. These sources differ significantly in spatial resolution and are influenced by variations in sensor parameters, solar irradiance, and atmospheric transmittance. This cross-platform, multiscale heterogeneity may result in feature distribution shifts. Accordingly, the use of such a diverse dataset enables a more rigorous evaluation of model generalization in complex real-world scenarios and provides an objective benchmark for model performance assessment.

It should be noted that the current analysis is limited to synthesized RGB imagery and does not include validation on raw multispectral or SAR data. Although MHNet employs a MAE mechanism, along with dropout and batch normalization, to improve generalization, effective adaptation to multispectral or SAR inputs would require further architectural modifications. Future research will focus on developing dedicated masked models tailored for multimodal remote sensing data to better accommodate the characteristics of different sensor types.

B. Evaluation Metrics

To evaluate the performance of the proposed method, four standard metrics commonly used in semantic segmentation tasks were adopted, as described below.

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1) Mean Intersection Over Union: The mean intersection over union (mIoU) metric measures the overlap between the predicted segmentation and the ground truth, expressed as the ratio of their intersection to their union. It is computed as

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{|P_i \cap G_i|}{|P_i \cup G_i|} \tag{9}$$

where N is the number of categories, i denotes the current category, P_i represents the predicted segmentation for that category, and G_i is the corresponding ground truth.

2) Recall: Recall quantifies the proportion of actual positive regions that are correctly identified by the model. It is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

where TP represents true positives and FN denotes false negatives.

3) Precision: Precision reflects the proportion of correct positive predictions among all predicted positive regions. It is given by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

where FP represents false positives.

4) F1 Score: The $F1$ score is the harmonic mean of precision and recall, providing a balanced evaluation of the two metrics. It is expressed as

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{12}$$

These metrics collectively provide a comprehensive evaluation of the segmentation model's performance, ensuring that it achieves both high detection rates and accurate predictions in lake water body segmentation tasks.

C. Implementation Details

The experimental hardware environment is an Intel[1] Xeon[1] Silver 4214R CPU, 128 GB memory, and NVIDIA GeForce RTX 3090 graphics card. The computational tasks were carried out using the PyTorch deep learning framework.

For optimization, the stochastic gradient descent (SGD) algorithm was employed, with hyperparameters kept at their default settings for this study. The training process utilized four worker threads and a batch size of 16. An initial learning rate of 0.001 was applied, following a cosine annealing scheduler that reduced the learning rate to a minimum of 0.00001. The momentum and weight decay parameters were set to 0.9 and 0.0001, respectively. The learning rate decreased by a factor of 0.1 during each scheduling cycle.

D. Results and Comparisons

In this section, we compare our proposed MHNet model with several representative methods for water body extraction, including PSPNet [43], DeepLabV3 [8], MultiResUNet [51], ResUNet++ [52], CENet [53], UNet++ [54], ResUNet-a [55], LANet [56], MANet [57], TransUNet [58], DFSNet [59], MSLWENet [30], SiameseNet-CB [31], DensePPMUNet-a [33], GAC-ViT [43], cross-city matters [60], SpectralGPT [61], and SeaMo [62]. The comparison was conducted on five datasets: TPD, GID, WHDLD, DLRSD, and ISIC2018 Dataset. Quantitative results are summarized in Tables I–V, respectively. The quantitative results of the comparative experiments are based on ten independent trials, with the mean and standard deviation calculated.

TABLE I
QUANTITATIVE EVALUATION ON TPD

Methods	mIoU(%)	Recall(%)	Precision(%)	F1(%)
PSPNet	90.74±0.73	95.07±1.21	95.17±0.63	94.99±0.42
DeepLabV3	89.36±0.23	94.52±0.30	94.21±0.26	94.28±0.13
MultiResUNet	92.71±0.54	92.56±0.70	93.23±1.02	95.11±0.30
ResUNet++	90.68±1.02	92.50±0.96	95.55±0.49	94.55±0.53
CENet	90.83±1.27	94.26±2.05	95.92±0.78	95.02±0.91
UNet++	93.49±0.28	95.96±0.47	97.25±0.36	96.49±0.16
ResUNet-a	93.27±0.55	95.55±0.65	97.40±0.26	96.36±0.30
LANet	90.92±0.76	93.97±0.96	96.46±0.37	95.07±0.43
MANet	84.80±0.26	86.56±0.29	96.30±0.39	91.61±0.16
TransUNet	93.06±0.42	95.72±0.63	97.01±0.34	96.25±0.24
DFSNet	89.25±0.56	94.11±1.58	95.54±0.57	93.51±0.45
MSLWENet	92.50±0.24	95.84±0.25	96.19±0.28	95.78±0.21
SiameseNet-CB	92.11±0.32	96.11±1.00	95.60±0.62	95.74±0.18
DensePPMUNet-a	90.83±0.45	95.16±0.86	95.13±0.77	95.01±0.26
GAC-ViT	91.98±0.49	94.84±0.39	96.28±0.19	95.44±0.14
Cross-city matters	94.19±0.54	96.29±0.43	97.57±0.35	96.60±0.13
SpectralGPT	93.44±0.35	96.37±0.53	97.11±0.55	96.71±0.15
SeaMo	92.91±0.43	96.13±0.51	95.48±0.54	96.17±0.21
Ours	**94.45**±0.43	**96.47**±0.62	**97.72**±0.26	**97.03**±0.24

TABLE II
QUANTITATIVE EVALUATION ON GID

Methods	mIoU(%)	Recall(%)	Precision(%)	F1(%)
PSPNet	49.81±3.14	72.12±2.69	53.45±3.78	57.99±2.52
DeepLabV3	52.84±1.61	76.21±0.58	57.31±1.70	67.71±1.07
MultiResUNet	61.86±2.30	74.86±2.49	66.00±3.16	67.60±1.56
ResUNet++	40.95±1.10	50.36±0.63	41.32±0.68	43.74±1.20
CENet	45.64±3.78	72.29±2.17	47.03±2.07	54.33±2.49
UNet++	64.70±2.13	76.27±1.57	68.73±2.70	69.97±1.44
ResUNet-a	61.91±0.71	75.36±2.38	65.53±1.50	67.49±0.52
LANet	48.69±2.91	67.96±3.42	54.39±3.85	57.25±2.17
MANet	43.41±2.01	70.42±2.02	51.67±3.04	57.36±1.93
TransUNet	58.59±1.71	73.03±2.02	63.07±2.36	65.07±1.21
DFSNet	48.91±0.99	66.19±4.25	53.57±0.72	58.16±0.61
MSLWENet	41.96±1.49	46.76±0.98	40.30±0.83	41.39±1.01
SiameseNet-CB	53.16±0.22	66.78±0.51	59.65±0.42	60.36±0.38
DensePPMUNet-a	51.00±6.56	70.03±2.45	55.87±8.35	59.06±4.86
GAC-ViT	64.64±1.07	78.29±0.81	61.94±2.11	66.91±1.30
Cross-city matters	67.23±0.39	**85.96**±0.13	73.92±0.33	77.73±0.38
SpectralGPT	61.72±0.13	78.77±0.81	70.43±0.35	74.36±0.41
SeaMo	64.18±0.18	84.01±0.37	69.75±0.56	76.46±0.59
Ours	**72.11**±0.37	82.03±0.67	**82.43**±0.16	**80.87**±0.32

On the TPD dataset, MHNet demonstrates clear advantages in both qualitative and quantitative analyses. Fig. 2 shows the visual comparisons, where MHNet accurately captures complex lake boundaries and even complex features such as water with bubbles. Quantitative results in Table I indicate that MHNet achieves the highest mIoU (94.45%), recall (96.47%), precision (97.72%), and $F1$-score (97.03%). These results surpass the second-best method by 0.26%, 0.10%, 0.15%, and 0.32%, respectively, highlighting its robustness and precision.

On the GID dataset, MHNet leads the second best method by 4.88%, 8.51%, and 3.14% in terms of mIoU, precision, and $F1$-score, respectively, as shown in Table II. This quantitative

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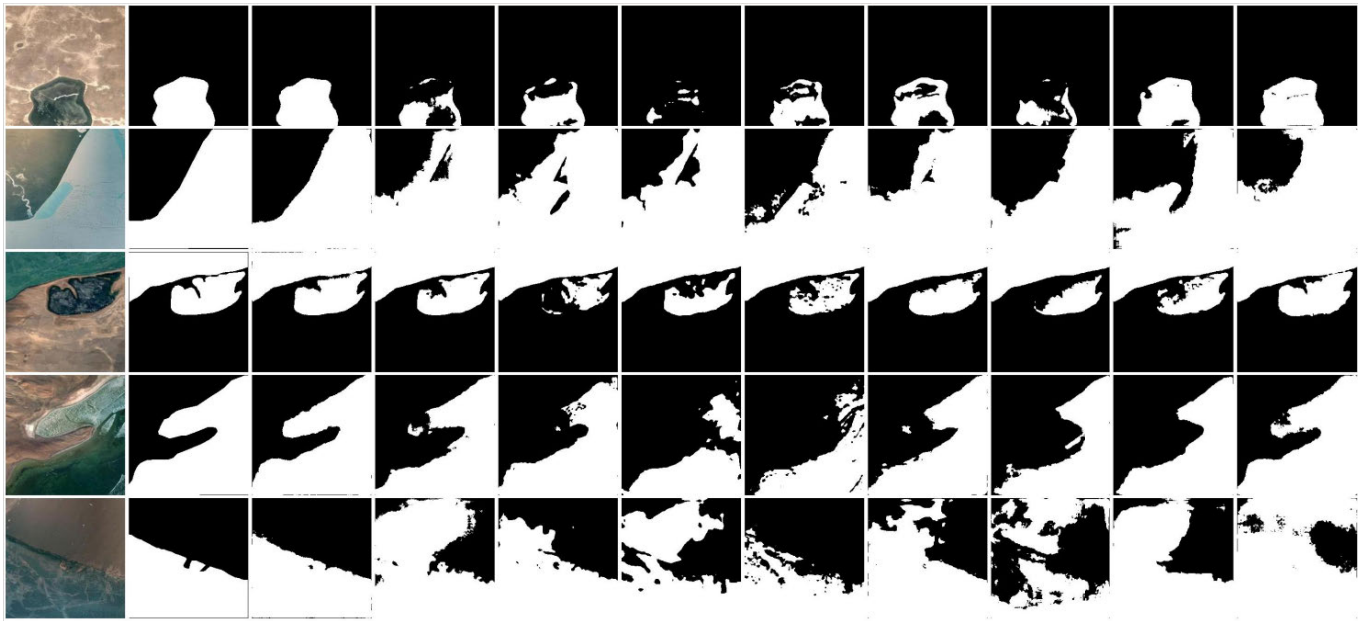


Fig. 2. Visualized segmentation results for several scenarios of dataset TPD: (a) original image, (b) ground truth, (c) our method, (d) DensePPMUNet-a, (e) TransUNet, (f) LANet, (g) MANet, (h) ResUNet-a, (i) CENet, (j) ResUNet++, and (k) MultiResUnet.

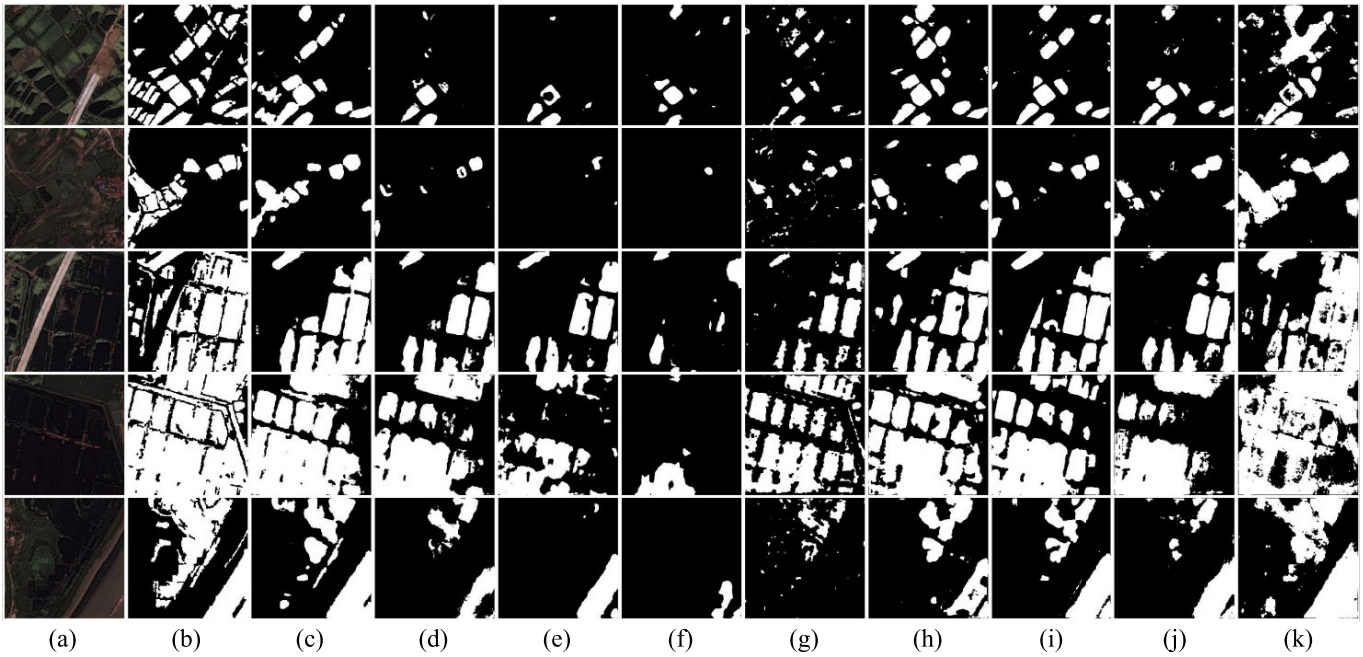


Fig. 3. Visualized segmentation results for several scenarios of dataset WHDLD: (a) Original image, (b) ground truth, (c) our method, (d) DensePPMUNet-a, (e) TransUNet, (f) LANet; (g) MANet, (h) ResUNet-a, (i) CENet, (j) ResUNet++, and (k) MultiResUnet.

result shows that MHNet is robust and advanced in water body extraction tasks.

On the WHDLD dataset, MHNet again outperforms all competing methods. Fig. 3 illustrates the qualitative results, where MHNet provides more accurate and detailed water segmentation. Table III presents the quantitative performance, showing improvements of 0.47%, 2.25%, and 1.08% in mIoU, precision, and $F1$-score over the second-best method. This reflects the model's adaptability to complex and densely labeled environments.

Since the DLRSD dataset was not involved in the construction of the training set, the evaluation on the DLRSD dataset is able to more adequately show the generalization performance of each method. Fig. 4 demonstrates that the segmentation results of MHNet are the closest to the ground truth. Specifically, MHNet can more completely extract water bodies such as ponds and lakes compared to the comparison method. Quantitative results in Table IV further confirm its effectiveness, with significant performance gains over other methods. This capability to generalize well to unseen data

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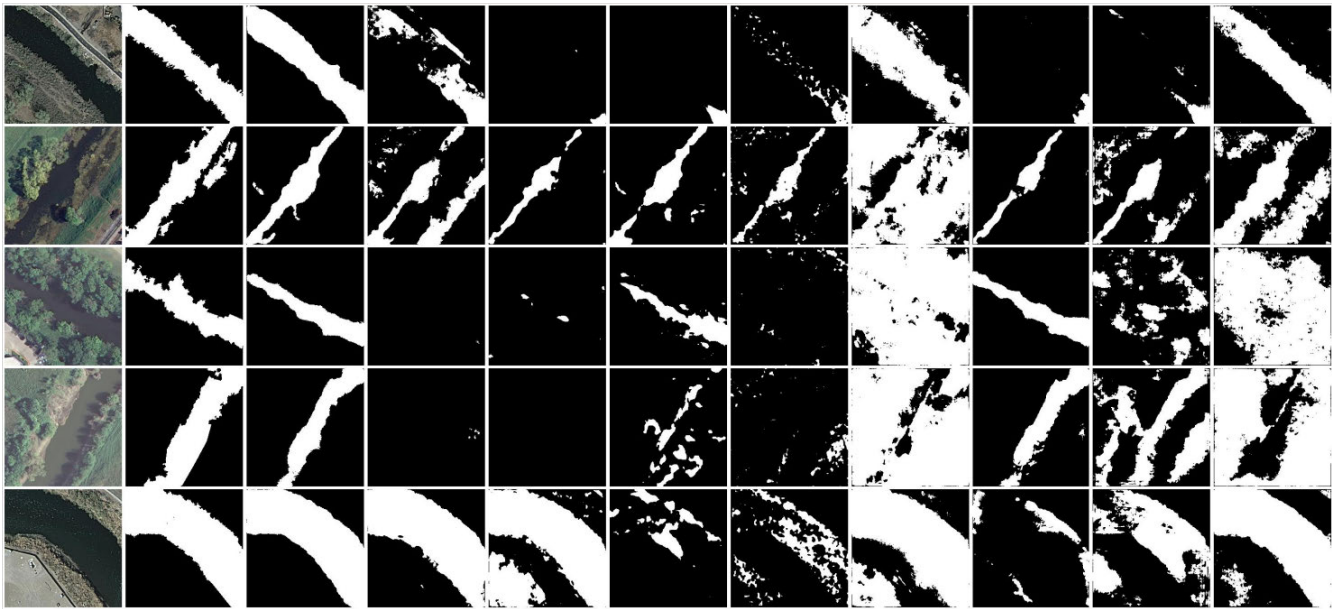


Fig. 4. Visualized segmentation results for several scenarios of dataset DLRSD: (a) original image, (b) ground truth, (c) our method, (d) DensePPMUNet-a, (e) TransUNet, (f) LANet, (g) MANet, (h) ResUNet-a, (i) CENet, (j) ResUNet++, and (k) MultiResUnet.

TABLE III
QUANTITATIVE EVALUATION ON WHDLD

Methods	mIoU(%)	Recall(%)	Precision(%)	F1(%)
PSPNet	73.16±5.50	79.95±2.96	89.29±7.66	83.60±4.12
DeepLabV3	69.56±1.10	86.55±0.36	78.68±1.42	82.66±0.65
MultiResUNet	78.75±2.51	89.61±1.11	91.58±3.60	87.77±1.58
ResUNet++	71.54±1.36	76.80±2.96	85.37±2.06	79.31±4.51
CENet	79.41±2.14	84.16±0.62	91.34±0.38	87.35±0.72
UNet++	81.23±2.14	85.15±3.58	93.65±1.95	89.03±1.30
ResUNet-a	77.93±1.26	86.77±1.97	87.62±1.11	86.78±0.77
LANet	70.34±7.32	76.75±9.38	90.00±7.78	81.52±5.50
MANet	63.94±1.75	71.41±2.98	85.54±1.09	77.51±1.48
TransUNet	80.92±2.22	89.72±0.91	89.03±3.13	88.94±1.46
DFSNet	64.89±1.24	69.42±2.56	91.25±0.98	75.56±0.91
MSLWENet	79.24±1.02	86.62±1.20	85.38±2.05	85.10±0.41
SiameseNet-CB	61.27±0.26	77.04±0.71	74.18±0.82	74.94±0.58
DensePPMUNet-a	77.62±2.03	85.63±2.97	89.05±5.47	86.84±1.29
GAC-ViT	81.78±1.22	87.34±0.88	91.73±0.41	88.21±0.97
Cross-city matters	85.43±0.36	**92.55**±0.43	92.80±0.37	91.22±0.37
SpectralGPT	83.65±0.48	85.95±0.19	94.80±0.78	90.93±0.59
SeaMo	81.47±0.13	83.77±0.27	94.66±0.22	89.60±0.66
Ours	**85.90**±0.17	89.83±0.13	**95.05**±0.11	**92.30**±0.10

TABLE IV
QUANTITATIVE EVALUATION ON DLRSD

Methods	mIoU(%)	Recall(%)	Precision(%)	F1(%)
PSPNet	42.34±3.27	48.17±4.50	75.68±5.41	57.44±3.55
DeepLabV3	37.71±2.77	43.04±6.12	77.26±8.44	54.08±2.36
MultiResUNet	36.61±3.90	48.82±8.00	58.76±7.50	51.26±4.03
ResUNet++	33.84±2.15	39.69±2.13	60.80±5.28	45.72±2.74
CENet	39.74±3.15	44.12±5.27	70.78±5.54	54.16±3.40
UNet++	36.24±5.83	41.23±7.71	75.69±8.10	51.44±6.20
ResUNet-a	33.87±2.47	36.40±2.79	79.95±7.99	48.61±3.18
LANet	41.38±5.88	47.90±6.71	70.75±7.73	56.08±6.47
MANet	31.29±0.82	32.41±1.22	74.94±0.63	44.60±1.04
TransUNet	44.24±3.55	47.43±5.11	85.28±4.36	59.70±3.13
DFSNet	41.27±2.23	46.24±2.10	74.76±3.72	54.49±1.97
MSLWENet	51.11±1.29	**59.56**±1.20	78.80±2.05	65.78±0.41
SiameseNet-CB	33.25±2.00	42.66±6.26	59.78±5.12	47.85±2.24
DensePPMUNet-a	46.13±5.30	47.50±8.90	75.50±7.06	61.14±5.37
GAC-ViT	38.90±1.70	47.98±2.53	74.35±2.21	57.84±3.34
Cross-city matters	46.03±1.23	48.87±0.37	80.32±1.63	60.08±1.36
SpectralGPT	43.93±1.24	53.91±1.38	66.49±1.17	58.49±0.51
SeaMo	45.23±1.41	57.68±0.59	68.46±1.38	61.56±0.38
Ours	**51.22**±0.83	53.38±0.97	**89.30**±0.36	**66.17**±0.66

TABLE V
QUANTITATIVE EVALUATION ON ISIC2018 DATASET

Methods	mIoU(%)	Recall(%)	Precision(%)	F1(%)
PSPNet	73.05±0.34	69.03±2.74	77.21±2.36	70.95±0.69
DeepLabV3	74.90±0.33	**80.17**±1.66	70.94±0.96	72.67±0.26
MultiResUNet	71.75±1.32	62.89±2.94	76.83±0.77	68.45±2.19
ResUNet++	74.64±0.80	70.95±3.46	77.47±1.76	73.00±1.45
CENet	74.42±0.82	73.10±3.14	76.49±2.74	72.97±0.96
UNet++	72.23±1.74	65.98±5.57	78.20±2.13	69.57±2.94
ResUNet-a	63.46±0.15	53.53±0.52	69.23±0.64	57.86±0.20
LANet	71.70±0.56	65.32±0.55	78.62±1.12	69.41±0.11
MANet	65.20±0.25	62.99±1.27	71.00±1.39	64.54±0.14
TransUNet	71.36±1.44	66.30±2.43	76.69±1.69	69.16±1.39
DFSNet	68.91±1.22	66.08±3.09	76.51±2.80	68.78±0.95
DensePPMUNet-a	75.16±1.55	70.19±3.17	79.10±0.13	72.14±1.89
GAC-ViT	67.63±0.75	73.36±1.07	65.19±1.50	67.21±0.38
Ours	**77.30**±0.18	75.01±0.59	**80.61**±0.28	**75.70**±0.25

is critical for real-world applications, where models often encounter conditions not represented in the training data.

On the ISIC2018 dataset, MHNet leads the second best method by 2.14%, 1.51%, and 2.70% in terms of mIoU, precision, and $F1$-score, respectively, as shown in Table V. Moreover, Fig. 5 shows the segmentation results of MHNet and comparative methods on the ISIC2018 dataset. MHNet achieves the segmentation map closest to the ground truth, outperforming other methods. This quantitative result shows that MHNet maintains a leading performance while ensuring generalization capability.

Across all datasets, MHNet consistently outperforms competing methods, achieving state-of-the-art results in both qualitative and quantitative evaluations. These findings validate the accuracy, robustness, and generalization capabilities of MHNet, making it a reliable solution for lake water body segmentation and similar tasks in remote sensing applications.

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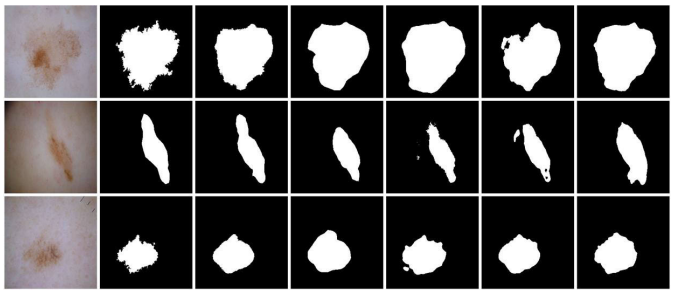


Fig. 5. Visualized segmentation results for several scenarios of ISIC2018 dataset: (a) original image, (b) ground truth, (c) our method, (d) PSPNet, (e) Deeplabv3, (f) UNet++, and (g) TransUNet.

TABLE VI
DIFFERENT MASKING RATIO ON THE TPD

Mask ratio	mIoU(%)	Recall(%)	Precision(%)	F1(%)
0.00	92.47	94.81	97.28	95.92
0.15	93.32	95.57	97.40	96.41
0.25	**94.45**	**96.47**	97.72	**97.03**
0.35	93.74	95.52	97.92	96.64
0.50	94.17	96.04	97.75	96.85
0.65	94.17	95.97	97.91	96.87
0.75	93.32	95.57	97.40	96.41
0.85	93.11	94.74	97.75	96.14

E. Ablation Studies

To comprehensively assess the efficacy of MHNet, we conducted a series of experiments focusing on critical architectural components, hyperparameters, and methodological choices. These studies aimed to identify the impact of different masking strategies, multiscale network depths, feature channel configurations, and essential modules like the convolutional embedding layer, Restormer blocks, and the MCFF module. In addition, we conducted experiments on the edge sharpness quantification, two hyperparameters of the loss function, the number of parameters and the computational complexity of MHNet at different multiscale depths and number of channels. The results of these ablation studies demonstrate how specific configurations enhance generalization ability, segmentation accuracy, and computational efficiency, ensuring that MHNet is robust across diverse datasets. Below, we detail the findings from our systematic exploration.

1) Masking Strategy: Masking strategies are a foundational aspect of MHNet, directly influencing the model's ability to generalize and learn meaningful representations. We analyzed the effects of masking ratios and masking methods to optimize this critical component.

a) Masking ratio: The masking ratio determines the extent of input information removed during training, encouraging the model to reconstruct missing details. As shown in Table VI, performance varies with the masking ratio, following a convex pattern. Masks that are scaled too low such as 0.15 perform poorly, mainly due to the lack of regularization they provide. And too high a ratio (e.g., 0.85) could result in insufficient information available for the reconstruction process, which could lead to performance degradation. A masking ratio of 0.25 strikes the optimal balance, achieving high accuracy while maintaining strong generalization.

TABLE VII
DIFFERENT MASKING METHOD ON THE TPD. RANDN USES RANDOM NOISE AS THE MASK, WHILE ZEROS USES ZERO AS THE MASK. FIXED MEANS THE SAME MASK RATES FOR THREE SCALES. UNIFORM MEANS THE SAME RANDOM RATE FOR ALL SCALES. RANDN MEANS THREE RANDOM RATES FOR SCALES

Method	mIoU(%)	Recall(%)	Precision(%)	F1(%)
Randn	**94.45**	**96.47**	97.72	**97.03**
Zeros	93.88	95.59	**97.97**	96.71
Fixed [0.25 0.25 0.25]	**94.45**	**96.47**	97.72	**97.03**
Fixed [0.25 0.5 0.75]	93.54	94.95	**98.19**	96.50
Uniform [0.1,0.9]	93.79	95.40	98.05	96.65
Randn [0.1,0.9]	93.81	95.49	98.04	96.68

TABLE VIII
DIFFERENT MULTISCALE DEPTH ON TPD. W/O MS MEANS BASELINE CONFIGURATION WITHOUT THE MULTISCALE MODULE

Depth	mIoU(%)	Recall(%)	Precision(%)	F1(%)
W/o ms	92.27	94.97	96.83	95.70
2	93.66	95.75	97.55	96.59
3	**94.45**	**96.47**	97.72	**97.03**
4	93.20	94.75	**98.14**	96.35

b) Masking method: First, we compared two masking methods: masking with zeros and masking with random noise. Random noise masking consistently outperformed zero masking, as shown in Table VII. This outcome suggests that random noise introduces more diverse patterns during training, helping the model learn robust features. Additionally, applying a uniform masking ratio of [0.25, 0.25, 0.25] across all scales outperformed varying ratios, as illustrated in Table VII. This finding underscores the importance of consistency in multiscale masking for achieving optimal performance. Thus, the final strategy employs random noise masking with a fixed ratio of 0.25 across three scales.

2) Multiscale Network Depth: Multiscale feature extraction is a key strength of MHNet. We evaluated the performance of the model with varying network depths (2, 3, and 4 levels) and a baseline configuration without the multiscale module. Table VIII shows that a depth of 3 achieves the best overall performance, outperforming both shallower and deeper configurations. A depth of 2 provides limited feature representation, while increasing to 4 introduces unnecessary complexity, leading to diminishing returns. This result aligns with the intuition that moderate depth balances feature diversity and computational feasibility. Consequently, the final model adopts a multiscale depth of 3.

3) Effects of the Essential Components: To validate the importance of key architectural components, we conducted experiments by selectively removing or replacing the convolutional embedding layer, Restormer blocks, MCFF module, and the mask module. The results are summarized in Table IX.

a) Convolutional embedding layer: Replacing the convolutional embedding layer with a FC embedding significantly increased computational overhead while sharply reducing the network performance. The convolutional embedding layer,

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TABLE IX

EXPERIMENTS ON CONV EMBEDDING AND MCFF MODULE IN MHNET ON THE FOUR DATASETS. W/O CE MEANS USING LINEAR EMBEDDING INSTEAD OF CONV EMBEDDING, W/O MCFF MEANS NOT USING MCFF, AND W/O MASK MEANS SET MASK RATIO TO ZERO

Method	Dataset	mIoU(%)	Recall(%)	Precision(%)	F1(%)
original	TPD	94.45	96.47	97.72	97.03
	GID	72.11	82.03	82.43	80.87
	WHDLD	85.90	89.83	95.05	92.30
	DLRSD	51.22	53.38	89.30	66.17
W/o CE	TPD	92.61	94.61	97.68	96.04
	GID	63.30	83.30	69.76	74.45
	WHDLD	67.44	81.67	95.05	92.30
	DLRSD	35.74	40.14	79.59	52.52
W/o MCFF	TPD	93.65	95.26	98.03	96.58
	GID	69.88	83.03	78.14	78.92
	WHDLD	85.73	89.22	95.52	92.20
	DLRSD	44.09	46.45	88.06	60.03
W/o Mask	TPD	92.47	94.81	97.28	95.92
	GID	59.90	67.44	68.03	65.31
	WHDLD	81.81	84.72	94.49	89.52
	DLRSD	50.27	52.24	87.70	64.53

TABLE X

DIFFERENT ATTENTION MECHANISM OF MHNET ON THE TPD

Method	mIoU(%)	Precision(%)	F1(%)	Params(M)	Flops(G)
Restormer blocks	94.45	97.72	97.03	26.60	14.99
MobileViT blocks	94.31	93.36	92.55	35.07	26.56

TABLE XI

DIFFERENT LOSS FUNCTION COMBINATION COEFFICIENTS ON THE TPD

α	β	mIoU(%)	Recall(%)	Precision(%)	F1(%)
0.50	1.00	**94.45**	**96.47**	97.72	**97.03**
1.00	0.50	94.28	96.22	97.76	96.93
0.50	0.50	92.65	94.06	98.21	96.03
0.50	1.50	94.10	95.81	98.00	96.83

TABLE XII

BOUNDARY $F1$-SCORE FOR MHNET AND OTHER METHODS ON THE FOUR EXPERIMENTAL WATER BODY SEGMENTATION DATASETS

Methods	TPD	GID	WHDLD	DLRSD
DeepLabV3	0.4802	0.1602	0.2730	0.2122
MANet	0.3923	0.0730	0.1816	0.1097
DFSNet	0.5353	0.2043	0.3309	0.2478
ResUNet-a	0.5586	0.2279	0.3582	0.2199
TransUNet	0.5461	0.2252	0.3687	0.2925
DensePPMUNet-a	0.5453	0.2544	0.4325	0.2864
GAC-ViT	0.4071	0.1183	0.1046	0.1787
ours	**0.6047**	**0.3151**	**0.4336**	**0.2988**

which allows for adjustable block sizes and overlaps via kernel size and stride, is more efficient and effective for processing remote sensing images.

b) MCFF module: The MCFF module facilitates multiscale feature fusion, which is important for accurate segmentation. Its removal caused a significant decline in performance, as shown in Table IX, underscoring its essential role in achieving robust multiscale integration.

c) Mask module: However, directly removing the mask module does not have as much impact on the evaluation metrics as the two modules mentioned above. The masking strategy enhances data efficiency and robustness while masking portions of the input and reconstructing them, the model develops a deeper understanding of the spatial and spectral relationships within the imagery, improving its generalization ability.

d) Restormer blocks: To verify the effectiveness of the attention mechanism in the Restormer blocks, we conducted ablation experiments. In the experiments, we replaced the attention mechanism of the Restormer blocks with those of other lightweight transformers, MobileViT and performed comparison tests on the same dataset, as shown in Table X. The results show that the attention mechanism of the Restormer module performs more significantly in terms of model performance improvement while maintaining a low computational overhead.

Overall, the above ablation studies provide valuable insights into the design of MHNet. The masking strategy not only enhances generalization but also demonstrates the nuanced impact of hyperparameters like masking ratio and method. Multiscale depth plays a pivotal role in balancing feature richness and computational cost, while the critical components—convolutional embedding, Restormer blocks, and MCFF—collectively ensure state-of-the-art performance in lake water segmentation.

These findings underscore the importance of meticulous architectural and methodological choices in developing models for complex remote sensing tasks. By isolating and analyzing individual elements, we ensure that MHNet is not only accurate but also generalizable and computationally efficient. This approach lays a strong foundation for future advancements and broader applications in water body extraction and beyond.

4) Hyperparameters in the Loss Function: The hyperparameters α and β in the loss function significantly influence model performance. We conducted ablation experiments to explore the impact of α and β, where α weights the BCELoss and β weights the dice loss. The experimental results are shown in Table XI. Testing various combinations, we found that $\alpha = 0.5$ and $\beta = 1.0$ achieved the best results, balancing pixel-level accuracy and spatial coherence. This setup effectively addresses class imbalance in remote sensing images and captures water body structures, especially in irregular regions. Experiments showed that improper α or β values lead to blurred boundaries or poor small-object extraction. Thus, the optimal configuration enhances model performance in water body segmentation tasks.

5) Edge Sharpness Quantification: In order to quantify the occasional oversmoothing problem in segmentation maps, we introduced the boundary $F1$ score [63] and the results are shown in Table XII. The boundary $F1$ score is a segmentation edge-based metric that evaluates how well the predicted boundary of an object matches the ground truth boundary. The results in Table XII show that MHNet achieved the lead on all four experimental waterbody segmentation datasets.

6) Model Parameters and Computational Overhead: Understanding the trade-off between model complexity and

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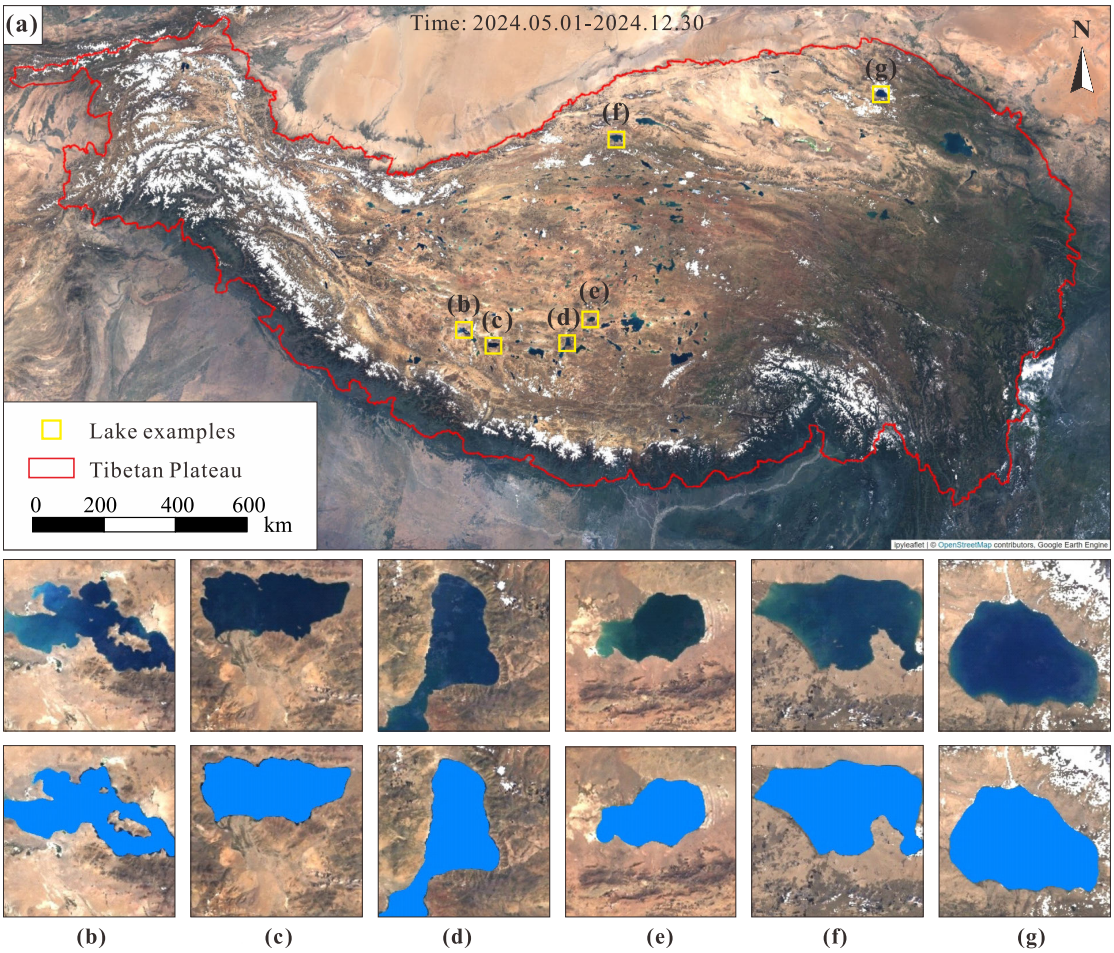


Fig. 6. Visualized water body segmentation results of MHNet and GEE integrated platform in Tibetan Plateau region. (a) Tibetan Plateau region from GEE. (b) Ngangla Ringco. (c) Taro co. (d) Tangra Yumco. (e) Daze Cuo. (f) Aqqikkol Hu. (g) Har Hu.

TABLE XIII

MODEL EFFICIENCY OF APPROACHES ON TPD, INCLUDING MHNET USING DIFFERENT FEATURE CHANNELS AND DEPTHS

Depth	Channels	mIoU(%)	Recall(%)	Params(M)	Flops(G)
2	64	93.66	95.75	2.37	6.36
	96	94.15	96.29	5.29	14.09
3	32	93.20	94.88	3.84	6.69
	64	**94.45**	**96.47**	26.60	14.99
4	32	93.19	95.26	82.97	9.48
	64	93.20	94.75	331.57	37.53

TABLE XIV

COMPARATIVE ANALYSIS OF COMPUTATIONAL OVERHEAD FOR MHNET AND OTHER METHODS, INCLUDING PARAMETER COUNTS, FLOPS, AND INFERENCE SPEED FOR EACH METHODS

Methods	Params(M)	Flops(G)	Inference time(ms)
DeepLabV3	5.81	**6.6**	**0.97**
MANet	44.38	36.6	4.09
DFSNet	36.39	29.19	2.01
ResUNet++	4.06	15.85	2.94
TransUNet	105.32	33.4	5.29
DensePPMUNet-a	**2.87**	63.42	8.82
GAC-ViT	90.24	835.8	42.23
ours	26.6	14.99	4.10

performance is important for practical deployment. We analyzed the impact of varying multiscale depths and feature channel numbers on parameter counts and computational overhead. Table XIII highlights that the configuration with a depth of 3 and 64 feature channels achieves the best balance. While increasing the number of feature channels slightly improves performance, it also increases computational costs disproportionately. This trade-off emphasizes the need for carefully calibrated configurations to ensure scalability without sacrificing accuracy. As shown in Table XIV, MHNet drastically reduces the computational overhead and the number of parameters compared to TransUNet and GAC-ViT. At the same time, MHNet also performs well in terms of FLOP, which is only 14.99 GMac, which is a significant reduction from 33.4 GMac in TransUNet and 835.8 GMac in GAC-ViT. In addition, the inference time of MHNet is only 4.10 ms, which demonstrates a significant advantage of MHNet model deployment compared to TransUNet and GAC-ViT.

F. Model Deployment and Online Inference

Leveraging the advanced architectural design of MHNet and the model hosting and deployment toolchain provided by Google's Vertex AI Platform, we successfully integrated

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MHNet with GEE platform to achieve scalable and efficient inference. GEE is a powerful and comprehensive geospatial analytics platform with a massive repository of satellite imagery and geospatial data. By integrating MHNet with GEE, we were able to directly utilize the platform's capabilities to perform water body segmentation tasks.

To evaluate the effectiveness of the integrated system, we conducted segmentation experiments in the Tibetan Plateau region (Fig. 6). Specifically, we segmented RGB images synthesized from Landsat 8 B4, B3, and B2 bands acquired from May to December 2024, at a scale of 50 km and a resolution of 3840 × 2048, dividing them into 256 × 256 image blocks. The average inference latency per block after deployment on Vertex AI was 900 ms, and the total processing time for the full-resolution image was 121.5 s. Notably, the relatively high single-batch latency is due to the absence of GPU acceleration on the Vertex AI endpoint. Nevertheless, the total processing time remains within an acceptable range for full-resolution segmentation.

The results demonstrate that MHNet, deployed on Google Cloud and integrated with GEE, achieves accurate and efficient lake segmentation in the region. This integration highlights MHNet's scalability for large-scale satellite image segmentation, positioning it as an effective solution for processing massive geospatial datasets with both high accuracy and efficiency.

V. Conclusion

In this study, a novel masked hybrid network for water body segmentation, named MHNet, which integrates masking properties into a hybrid convolution-transformer architecture to address key challenges in the task. By leveraging a modified scaling interchannel masking fusion strategy, MHNet overcomes limitations associated with insufficient datasets and enhances the model's representation learning capabilities. Furthermore, the proposed MCFF module enables effective multiscale feature integration, significantly improving segmentation accuracy, particularly for complex water body boundaries. Extensive experiments on four public datasets demonstrate that MHNet outperforms state-of-the-art methods in terms of accuracy, generalization, and robustness. The combination of hybrid architecture, MAE mechanisms, and multiscale feature fusion positions MHNet as a robust and efficient solution for water body segmentation. The successful deployment and integration of MHNet with GEE demonstrates the model's adaptability and scalability in realistic scenarios. This work paves the way for further applications in environmental monitoring, disaster response, and other geospatial analysis tasks, taking full advantage of the combined strengths of advanced machine learning models and powerful geospatial platforms.

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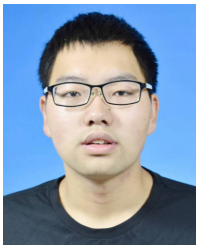
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